MITIGATA HACKATHON

Problem statements

Six challenges across security, data and product engineering.

THE SIX CHALLENGES

01 Kill chain and attack paths

02 Multi-tenant aggregation layer

03 Community scam map

04 Autonomous pentest engine

05 Config-driven integration platform

06 SAST and DAST correlation

Confidential to Mitigata

PROBLEM 01

Kill chain and attack path analysis

Analyse large volumes of security logs to detect a concealed attack, and present it as one narrative: where it began, what the attacker did, and which systems they can reach next.

Data tiers: 15, 30 or 50 GB of logs

01 Detect and label

Find suspicious events and tag each with a MITRE ATT&CK technique and kill chain stage.

02 Connect the steps

Link events across hosts, users, IPs and time into one attack path, without running out of memory.

03 Explain and look ahead

Show a timeline and a graph of hosts and users, plus what the attacker could reach next.

REAL INCIDENT · INC-01

A 2013 retail breach: one stolen vendor login, weeks of ignored alerts, up to 110 million customers.

PROBLEM 02

Multi-tenant aggregation layer

Build a layer between the database and the dashboards, so counts, trends and top-10 lists load fast, stay close to live, and never slow the main database.

Scale: 10M records, 100 tenants

01 **Fast and fresh**

Dashboard queries return in under a second, and new or updated orders show up within minutes.

02 **Four dashboard queries**

Counts and totals, trends over time, top-N lists and breakdowns by category, through a simple API.

03 **Separate and correct**

Each tenant sees only its own data, and the numbers match a direct database query.

REAL INCIDENT · INC-02

A 2023 cache bug that showed some users other users' chat titles and payment details.

PROBLEM 03

Community scam map

Build an app where people warn each other about scam phone numbers, UPI IDs and links, and check one before they trust it.

Reporters stay anonymous

01 **Report and look up**

Report a scam in under 60 seconds. Lookups answer in under half a second.

02 **Map and alerts**

Show the last 7 days of reports by area, and alert only when something is unusual.

03 **Beat fake reports**

Trust rules must stop report-bombing, fake accounts, spam and junk, without missing real scams.

REAL INCIDENT · INC-03

India's "digital arrest" scams: over 83,000 WhatsApp accounts blocked by March 2025.

PROBLEM 04

Autonomous pentest engine (CART)

Build a team of agents that map a hidden network's web, network and binary attack surface, prove its weaknesses, and write one report following PTES.

Test only inside the given scope

01 **Discover all three surfaces**

Find live hosts, ports and services, then crawl web apps and inspect the binaries.

02 **Coordinate agents**

Agents share findings and hand work to each other, so no host is tested twice.

03 **Verify and report**

Prove each weakness, tag it with a PTES phase, and deliver one report with the attack path.

REAL INCIDENT · INC-04

A 2019 bank breach: a misconfigured web firewall led to cloud data on about 100 million people.

PROBLEM 05

Config-driven integration platform

Build a platform where connecting a new data source, such as a firewall, Okta or AWS, means writing a config file, not new code.

Prove it with 3 different sources

01 Config describes the source

Login, endpoints, pagination, schedule, and a mapping into one common format.

02 Platform handles the rest

Safe secrets, retries with backoff, rate limits and bad data, the same for every connector.

03 Health stays visible

A simple page shows each connection's status, last run and errors.

REAL INCIDENT · INC-05

A 2025 breach where stolen integration tokens exposed many companies' CRM data.

PROBLEM 06 FLAGSHIP

SAST and DAST correlation

Scan an app's source code and its running version, then connect the results so each real vulnerability appears once, with the line of code and proof it can be exploited.

Solve it for bonus points

01 Scan code and app

Cover at least four of six vulnerability types, including SQL injection and XSS, in one common format with CWE numbers.

02 Correlate and merge

Link a runtime finding to the code behind it and merge duplicates. This is what we judge most.

03 Prioritise what to fix

Label each issue Confirmed, Code only or Runtime only, and rank confirmed issues first.

REAL INCIDENT · INC-06

A 2017 credit bureau breach: a known web flaw that scans missed, about 145 million people.